## 5.1 General Guidelines

- Obstetrical OB patients > 20 weeks gestation with a primary OB diagnosis should be treated as per the Obstetrical Transport protocols.
- Patients who require brief, deep, dissociative, or anesthetic level sedation should be treated as per the Procedural Sedation protocol.
- Intubated patients should be treated as per the Post Intubation Sedation protocol.
- Established SpO2, cardiac rhythm, NIBP, and respiratory rate monitoring prior to therapy.
- Continuous non-invasive waveform capnography must be monitored if multiple doses of any single medication are administered, or if opioids and benzodiazepines are used concomitantly.
- Administer analgesics and sedatives in lower doses, over longer duration, and at increased intervals if vital signs, medications given prior to arrival, clinical problem, age, comorbidities, hemodynamic status, or the presence of Intoxicants indicate it.
- Approved pain and sedation assessments should be to set therapeutic goals, assess patient response and document pain and sedation.
- General approach to drug delivery
    1. Select medication and dose based on clinical considerations
    2. Initially, use recurrent medication boluses at short intervals until goal is attained
    3. Use scheduled boluses at longer intervals to maintain goal
    4. Anticipate pain and anxiety during transport and treat preemptively
    5. Re-bolus as needed
- Analgesics and anxiolytics given IV push should be administered slowly over 1-2 minutes


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## 5.1.1 Pain and Sedation Scales

### Critical-Care Pain Observation Tool (CPOT)

| Indicator | Description | Score | |
|---|---|---|---|
| Facial Expression | No muscular tension observed | Relaxed/neutral | 0 |
| | Frowning, brow lowering, orbit tightening, and levator contraction | Tense | 1 |
| | All facial movements above, plus eyelid tightly closed | Grimacing | 2 |
| Body Movements | Does not move at all (does not necessarily mean absence of pain) | Absence of movements | 0 |
| | Slow, cautious movements, touching or rubbing pain site, seeking attention through movements | Protection | 1 |
| | Pulling tube, attempting to sit up, moving limbs, thrashing, not following commands, striking at staff, trying to climb out of bed | Restlessness | 2 |
| Muscle Tension | No resistance to passive movements | Relaxed | 0 |
| | Resistance to passive movements | Tense, rigid | 1 |
| | Strong resistance to passive movements, inability to complete them | Very tense or rigid | 2 |
| **Only score compliance with ventilator or vocalization depending on patient status** | | | |
| Ventilator Compliance (intubated patients) | Alarms not activated, easy ventilation | Tolerating ventilator or movement | 0 |
| | Alarms stop spontaneously | Coughing but tolerating | 1 |
| | Asynchrony: blocking ventilation, alarms frequently activated | Fighting ventilator | 2 |
| Vocalization (non-intubated patients) | Talking in normal tone or no sound | Talking in normal tone or no sound | 0 |
| | Sighing, moaning | Sighing, moaning | 1 |
| | Crying out, sobbing | Crying out, sobbing | 2 |

NOTE: **Select 1 score for each category and total**
NOTE: **Score Range: 0 (little/no pain) to 8 (severe pain)**

### Behavioral Pain Scale (BPS)

| Item | Description | Score |
|---|---|---|
| Facial | Relaxed | 1 |
| | Partially tightened (e.g., brow lowering) | 2 |
| | Fully tightened (e.g., eyelid closing) | 3 |
| | Grimacing | 4 |
| Upper Limbs | No movement | 1 |
| | Partially bent | 2 |
| | Fully bent with finger flexion | 3 |
| | Permanently retracted | 4 |
| Compliance With Ventilator | Tolerating movement | 1 |
| | Coughing but tolerates ventilator most of the time | 2 |
| | Fighting ventilator | 3 |
| | Unable to control ventilation | 4 |

NOTE: **Select 1 score from each category and total**
NOTE: **Score range 3 (little/no pain) to 12 (severe pain)**

## Richmond Agitation Sedation Score (RASS)

| Score | Term | Description |
|---|---|---|
| 4 | Combative | Combative or violent, immediate danger to staff |
| 3 | Very agitated | Pulls on or removing catheters/tubes or aggressive behavior |
| 2 | Agitated | Frequent nonpurposeful movement or ventilator dyssynchrony |
| 1 | Restless | Anxious but movements not aggressive or vigorous |
| 0 | Alert and calm | |
| -1 | Drowsy | Not fully alert, >10 secs awakening, with eye contact, to voice |
| -2 | Light sedation | Briefly (< 10 secs) awakens with eye contact to voice |
| -3 | Moderate sedation | Any movement (but no eye contact) to voice |
| -4 | Deep sedation | No response to voice, any movement to physical stimulation |
| -5 | Unarousable | No response to voice or physical stimulation |

## Riker Sedation-Agitation Scale (SAS)

| Score | Term | Descriptor |
|---|---|---|
| 7 | Dangerous Agitation | Pulling at ET tube, trying to get off cot, striking at crew, thrashing side-to-side |
| 6 | Very Agitated | Requiring restraint and frequent verbal reminding of limits, biting ETT |
| 5 | Agitated | Anxious or physically agitated, calms to verbal instructions |
| 4 | Calm and Cooperative | Calm, easily arousable, follows commands |
| 3 | Sedated | Difficult to arouse but awakens to verbal or physical stimuli, follows commands but drifts off |
| 2 | Very Sedated | Arouses to physical stimuli but does not communicate or follow commands, may move spontaneously |
| 1 | Unarousable | Minimal or no response to noxious stimuli, does not communicate or follow commands |


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## 5.1.2 Analgesia

Assess pain frequently during transport to gauge severity and monitor the response to interventions. Pain assessment should include the use of scales appropriate for the age group and patient condition. Pain should be treated to a level as low as reasonably possible in balance with the patient's cardiovascular stability and pain goals

### Fentanyl

**Adult**

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN

**Pediatric**

- Refer to adult dosing

### Morphine

**Adult**

- 2-10 mg IV push over 1-2 min, IM
- Repeat every 20 minutes PRN
- Titrate to pain goal

**Pediatric**

- 0.05-0.2 mg/kg IV over 1-2 min, IM
- Max single dose 10 mg
- Repeat every 10 minutes PRN
- Titrate to pain goal

### Hydromorphone

**Adult**

- 0.2–1 mg IV over 2 min, IM
- Repeat q 15 min PRN
- Max total dose 4 mg

**Pediatric**

- 0.01–0.02 mg/kg IV over 2 min, IM
- Repeat q 15 min PRN
- Max single dose 1 mg
- Max total dose 4 mg

## Adjunctive Analgesia

### Ketamine—Sub-dissociative Dose

**Adult**

- 0.1–0.3 mg/kg IBW IV over 1-2 min, IM, NAS
- Repeat q 10–20 min PRN
- Max single dose 30 mg

**Pediatric**

- Refer to adult dosing

NOTE: **For pain refractory to opioid analgesia**

NOTE: **Ketamine should be used for patients on Naltrexone or Buprenorphine**

NOTE: **Sub-dissociative doses of ketamine is not advised in patients experiencing psychosis**

⚠ CAUTION **Sub-dissociative doses of ketamine for pain control must be given slowly. If not, partial dissociation can occur, which is very disturbing to the patient and can result in agitation and delirium.**

### Ketorolac

**Adult**

- 15 mg IV over 30–60 seconds

**Pediatric**

- 0.5 mg/kg IV over 30–60 seconds
- Max dose 15 mg

## 5.1.3 Therapeutic Opioid Reversal

### Naloxone—Therapeutic Opioid Respiratory Depression

**Adult**

- 0.001–0.01 mg/kg IV/IM/IN
- Repeat every 2–10 min PRN

**Pediatric**

- Refer to adult dosing

NOTE: **Reversal of respiratory depression from therapeutic opioid dosing**

NOTE: **Titrate to resolve opioid induced respiratory depression**

### Naloxone—Advisories

- Use with caution in chronic opiate users
- Airway protection and supportive measures may be preferable to reversal
- Naloxone duration of action may be shorter than the opiate being reversed
- IV administration is preferred
- Use a mucosal atomizer device (MAD) for intra-nasal administration


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## 5.1.4 Anxiolysis

- Treat anxiety with titrated benzodiazepines after pain has been controlled with opioids.
- Benzodiazepines used in combination with opioids should be administered with extreme caution and careful patient monitoring.
- Anxious patients can escalate to Dangerous Agitation at any time. Therefore, preparation should be made to respond to Dangerous Agitation.

### Indications

- Self reported anxiety
- Evidence of anxious affect

### Management

- Treat prior to loading in aircraft if possible

### Medications

#### Midazolam

**Adult**

- 0.5–5 mg IV over 1-2 min, IM, NAS
- Repeat every 5 min PRN

**Pediatric**

- 0.05–0.1 mg/kg IBW IV over 1-2 min, IM, NAS
- Maximum single dose 5 mg
- Repeat every 2 min PRN

#### Lorazepam

**Adult**

- 0.5 to 2 mg IV over 1 min, IM, NAS
- Repeat every 10-20 min PRN

**Pediatric**

- 0.05–0.1 mg/kg IV over 1 min, IM, NAS
- Maximum single dose 2 mg
- Repeat every 10-20 min PRN


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## 5.2 Procedural Sedation

Procedural sedation may be used for short term tolerance of unpleasant procedures.

- Patients who require analgesia or anxiolysis should be treated as per the Analgesia and Anxiolysis Guideline
- Pulse oximetry (SpO2), cardiac rhythm, noninvasive blood pressure (NIBP), continuous waveform capnography and respiratory rate monitoring must be established prior to initiating procedural sedation.
- Equipment and medications to manage oxygenation, ventilation, and CV instability should be prepared and ready prior to initiating procedural sedation.
- CPR induced consciousness

**⚠ CAUTION** **Patients who receive dissociative ketamine doses should be monitored closely for apnea and airway compromise. Medical Crews must always be prepared to manage the airway.**

### 5.2.1 Short Term Painful or Unpleasant Procedures

Procedural sedation may be used for the following short term painful and unpleasant procedures:

- Cardioversion
- Tube or simple thoracostomy
- Fracture realignment and splinting
- Extrication and disentanglement

#### Ketamine

**Adult**

- 1-2 mg/kg IBW IV over 30–60 seconds
- 4 mg/kg IBW IM

**Pediatric**

- Refer to adult dosing

**NOTE:** **Ketamine is the preferred medication for procedural sedation**

**NOTE:** ~~**Use lower dose in patients with cardiovascular instability**~~

#### Etomidate

**Adult**

- 0.1 mg/kg TBW IVP over 30–60 sec
- Repeat 0.05 mg/kg TBW every 3–5 minutes PRN
- Maximum single dose 30 mg
- Pretreat with ondansetron PRN

**Pediatric**

- 0.1 mg/kg TBW IVP over 30-60 sec
- Repeat 0.05 mg/kg TBW every 3-5 minutes PRN
- Maximum single dose 20 mg
- Pretreat with ondansetron PRN

**NOTE:** **Dose calculation should be based on total body weight.**

**NOTE:** **Etomidate may be used as an alternate to ketamine for very brief procedures.**

#### Etomidate—Advisories

- Use with caution in patients with acute adrenal insufficiency
- Do Not use for prolonged or post intubation sedation

**⚠ CAUTION** **While myoclonus is more likely after rapid IV push etomidate, it may occur regardless of the medication delivery method. Crewmembers and bystanders need to be aware that quick, involuntary muscle jerks may occur after administration.**

## 5.3 Post Intubation Analgesia and Sedation

Post intubation analgesia and sedation should be carefully delivered based on the patients condition and assessment findings.

- Non Intubated patients who require brief, deep, disassociative, or anesthetic level sedation should be treated as per the procedural sedation guideline.
- Non Intubated patients who require treatment for pain and anxiety should be treated as per the analgesia and anxiolysis guideline.
- Continuous waveform capnography, SpO2, cardiac rhythm, NIBP, and respiratory rate monioring must be established at the time of rendezvous.
- Administer analgesics and sedatives in lower doses, over longer duration, and at increased intervals if vital signs, medications given prior to arrival, clinical problem, age, comorbidities, hemodynamic status, or the presence of Intoxicants indicate it.
- Therapeutic goals should be established prior to initiating pharmacologic interventions.
- Pain and agitation should be assessed and documented using the RASS, SAS, CPOT, or BPS validated assesment tools.

NOTICE

**Patients who have received NMBs or otherwise cannot be assessed should receive empiric analgesia and sedation using targeted dose goals.**

### 5.3.1 General Approach to Drug Delivery

- Select dose based on assessment and clinical considerations
- Initially use repeat bolus administration until the pain and sedation goal is attained
- Check vital signs between each bolus and adjust dose PRN
- After initial goal is attained use scheduled bolus or infusion to maintain goal
- Anticipate pain and agitation during transport and treat preemptively by bolus
- Treat break through agitation or pain by bolus medication PRN
- IV push analgesics and sedatives should be administered slowly over 1-2 min.
- Sedation goal for this guideline is a RASS -1 to -5

### 5.3.2 Controlled Substance Infusions

- Taking possession of a controlled substance infusion established by another provider or facility is prohibited.
- Transferring possession or leaving behind a partially used PHI controlled substance infusion is prohibited.
- If a controlled substance infusion established by a sending provider is to be continued, a new infusion must be prepared using PHI controlled substances.

#### Residual Controlled Substance Infusion Disposal

- Residual or partly used controlled substance infusions must be accurately accounted for and wasted after patient hand off.
- See Controlled Substance Disposal for more information

NOTICE

**Taking or transferring possession of controlled substances is prohibited. If a controlled substance infusion is required during transport, it should be prepared using PHI controlled substances. Unused controlled substances must be wasted and documented at the end of the transport.**

## Patient Stratification

Stratify the patient using the following categories and apply the applicable part of the guideline:

- Patients whose level of sedation and response to medications **CAN** be assessed and used to guide therapy
- Patients given neuromuscular blockers or whose level of sedation and response to medications **CANNOT** be assessed
- Patients with hemodynamic instability e.g. hypotension, tachycardia, shock index ≥ 1.

## Able to Assess Pain and Agitation

- Assess pain and agitation and using an approved self-reporting or observational scale
- Establish a treatment goal
- Use an opioid analgesic or dissociative dose of ketamine
- Use one agent exclusively until it's use and application has been optimized

**⚠ CAUTION** ***With the EXCEPTION of Ketamine, use the lower part of the dose range or reduce the normal dose by 1/2 to 1/4 of sedatives if the patient is hemodynamically unstable.***

### Fentanyl

**Adult**

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN
- Titrate to sedation goal

Alternatively

- 1-2 mcg/kg/hr via infusion pump

**Pediatric**

- Refer to adult dosing

### Ketamine

**Adult**

- 1–2 mg/kg IBW IV over 30–60 seconds
- Repeat bolus dose every 10–20 minutes to achieve the desired sedation goal

To maintain post-Intubation sedation goal after initial IV bolus therapy

- Repeat IV bolus every 20–30 minutes, OR
- 1–4 mg/kg/hr continuous IV infusion

**Pediatric**

- Refer to adult dosing

**NOTE:** **Ketamine is the preferred medication for post-intubation analgesia and sedation if the patient is unstable.**

**⚠ WARNING** **Administer one or more initial bolus doses of 1-2 mg/kg IBW IV before starting a ketamine infusion of 1-4 mg/kg/hr. Failing to administer the bolus dose prior to the infusion can lead to under-sedation and dysphoria. The bolus ensures that a dissociative dose is achieved, which is critical for rapid and effective sedation.**

## Concomitant Adjunctive Sedatives

Continue to use fentanyl or ketamine as mono-therapy as long as pain is controlled and the sedation (RASS) goal is met and sustained. Sedatives may be added as a second adjunctive agent if RASS goals can not be maintained with fentaNYL or ketamine mono-therapy.

**NOTICE** **A Sedative should be added if opioids fully control pain but sedation goals cannot be met. Choose only 1 sedative to use.**

### Propofol—Post Intubation Agitation

**Adult**

- 5–100 mcg/kg/min via infusion pump
- Titrate 5–10 mcg/kg/min ~ every 5 min to sedation goal
- Doses greater than 100 mcg/kg/min require a physician order

**After infusion is initiated**

- 0.5–1 mg/kg IV bolus may be used to supplement infusion
- May repeat every 1 to 3 minutes PRN

**Pediatric**

- Refer to adult dosing

NOTE: **If a sedation assessment is not possible, an empiric target dose of ~50 mcg/kg/min should be used**

NOTE: **Bolus dosing must be used concomitantly with an infusion**

### Propofol—Advisories

NOTE: **Medical Crewmembers must administer Propofol in compliance with their scope of practice.**

- Continue infusions in place at patient rendezvous using the existing dose and titrate PRN
- Propofol lacks analgesic properties; therefore, pain management necessitates the concomitant administration of an opioid.
- Effective dosages in the hospital unit may provide insufficient sedation during transport due to stimuli and pain.
- Resuscitative fluids or vasopressors may be required during propofol administration to prevent hypotension.

### Midazolam

**Adult**

- 0.025 to 0.05 mg/kg IV
- Repeat every 10 min PRN to achieve the sedation goal
- Repeat PRN for breakthrough agitation

To maintain goal

- Repeat bolus every 20–60 min, OR
- 0.05 to 0.2 mg/kg/hr continuous infusion

**Pediatric**

- Refer to adult dosing

NOTE: **Midazolam is the preferred sedative agent if there is a likelihood of seizure activity masked by NMB.**

### Lorazepam

**Adult**

- 0.025-0.1 mg/kg IV over 1 min
- Maximum single dose 2 mg
- Repeat every 10-20 min PRN
- May use if midazolam is not available

**Pediatric**

- Refer to adult dosing

### Dexmedetomidine

**Adult**

- 0.2–1.5 mcg/kg/hour IV infusion
- Titrate by 0.2 mcg/kg/hour every 30 minutes to sedation goal

**Pediatric**

Refer to adult dosing

## Unable to Assess Level of Pain or Sedation

Patient who have received neuromuscular blocking agents or are otherwise unable to be assessed for pain and agitation should be empirically treated using Ketamine mono-therapy or an opioid plus sedative dual agent combination.

**⚠WARNING** **Pain and sedation scales such as CPOT, BPS, SAS and RASS, measure behavors that a paralyzed patient cannot display. These scores can not be used to assess and direct analgesia and anxiolysis in patient that have received NMBs.**

**⚠CAUTION** **Multiple or alternating therapeutic approaches may cause adverse events. One pain/sedation approach should be chosen, and then optimized.**

### Ketamine Mono-therapy

#### Ketamine

**Adult**

- 1–2 mg/kg IBW IV over 30–60 seconds
- Repeat bolus dose every 10–20 minutes to achieve the desired sedation goal

To maintain post-Intubation sedation goal after initial IV bolus therapy

- Repeat IV bolus every 20–30 minutes, OR
- 1–4 mg/kg/hr continuous IV infusion

**Pediatric**

- Refer to adult dosing

NOTE: **Ketamine is the preferred medication for post-intubation analgesia and sedation if the patient is unstable.**

**⚠WARNING** **Administer one or more initial bolus doses of 1-2 mg/kg IBW IV before starting a ketamine infusion of 1-4 mg/kg/hr. Failing to administer the bolus dose prior to the infusion can lead to under-sedation and dysphoria. The bolus ensures that a dissociative dose is achieved, which is critical for rapid and effective sedation.**

## Opioid PLUS Sedative Dual Agent Therapy

NOTE: **Administer fentanyl before administering concomitant sedative**

#### Fentanyl

**Adult**

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN
- Titrate to sedation goal

Alternatively

- 1-2 mcg/kg/hr via infusion pump

**Pediatric**

- Refer to adult dosing

## Concomitant Sedatives

**Choose one**

### Propofol

**Adult**

- 5–100 mcg/kg/min via infusion pump
- Titrate 5–10 mcg/kg/min ~ every 5 min to sedation goal
- Doses greater than 100 mcg/kg/min require a physician order

**After infusion is initiated**

- 0.5–1 mg/kg IV bolus may be used to supplement infusion
- May repeat every 1 to 3 minutes PRN

NOTE: **If a sedation assessment is not possible, an empiric target dose of ~50 mcg/kg/min should be used**

NOTE: **Bolus dosing must be used concomitantly with an infusion**

**Pediatric**

- Refer to adult dosing

### Midazolam

**Adult**

- 0.025 to 0.05 mg/kg IV
- Repeat every 10 min PRN to achieve the sedation goal
- Repeat PRN for breakthrough agitation

To maintain goal

- Repeat bolus every 20–60 min, OR
- 0.05 to 0.2 mg/kg/hr continuous infusion

NOTE: **Midazolam is the preferred sedative agent if there is a likelihood of seizure activity masked by NMB.**

**Pediatric**

- Refer to adult dosing

## Post Intubation Pain and Agitation Medications

### Fentanyl

**Adult**

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN
- Titrate to sedation goal

Alternatively

- 1-2 mcg/kg/hr via infusion pump

**Pediatric**

- Refer to adult dosing

### Ketamine

**Adult**

- 1–2 mg/kg IBW IV over 30–60 seconds
- Repeat bolus dose every 10–20 minutes to achieve the desired sedation goal

To maintain post-Intubation sedation goal after initial IV bolus therapy

- Repeat IV bolus every 20–30 minutes, OR
- 1–4 mg/kg/hr continuous IV infusion

NOTE: **Ketamine is the preferred medication for post-intubation analgesia and sedation if the patient is unstable.**

**Pediatric**

- Refer to adult dosing

**Administer one or more initial bolus doses of 1-2 mg/kg IBW IV before starting a ketamine infusion of 1-4 mg/kg/hr. Failing to administer the bolus dose prior to the infusion can lead to under-sedation and dysphoria. The bolus ensures that a dissociative dose is achieved, which is critical for rapid and effective sedation.**

### Propofol

**Adult**

- 5–100 mcg/kg/min via infusion pump
- Titrate 5–10 mcg/kg/min ~ every 5 min to sedation goal
- Doses greater than 100 mcg/kg/min require a physician order

**After infusion is initiated**

- 0.5–1 mg/kg IV bolus may be used to supplement infusion
- May repeat every 1 to 3 minutes PRN

NOTE: **If a sedation assessment is not possible, an empiric target dose of ~50 mcg/kg/min should be used**

NOTE: **Bolus dosing must be used concomitantly with an infusion**

**Pediatric**

- Refer to adult dosing

### Midazolam

**Adult**

- 0.025 to 0.05 mg/kg IV
- Repeat every 10 min PRN to achieve the sedation goal
- Repeat PRN for breakthrough agitation

To maintain goal

- Repeat bolus every 20–60 min, OR
- 0.05 to 0.2 mg/kg/hr continuous infusion

NOTE: **Midazolam is the preferred sedative agent if there is a likelihood of seizure activity masked by NMB.**

**Pediatric**

- Refer to adult dosing